

‘Influência do Comportamento Sedentário no Desenvolvimento da Doença Hepática Gordurosa Associada à Disfunção Metabólica: Análise Exploratória com Dados do NHANES’

Guilherme Ferreira

2025-26-04 (YYYY-DD-MM)

Teste

Teste

Teste

Teste **Teste** - negrito *Teste* - Itálico

R Markdown

This code was wrote in r language as a R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document.

Here, variable names in CAPS are coded as in the original dataset (with the exception of missing value codes being set to NA and some cases assigned values based on skip patterns). Variable names in lowercase were derived from other variables (e.g., smoker).

Sites importantes

<https://wwwn.cdc.gov/nchs/nhanes/search/default.aspx> <https://www.epirhandbook.com/en/> https://wwwn.cdc.gov/nchs/data/tutorials/module3_examples_R.r <https://libstc.cc/#/stc/nid:6e7ka238xj6qcuyf7dkdkuyfo> <https://bookdown.org/rwnahhas/RMPH/intro.html>

1. Using Packages

```
rm(list = ls())

# Display using packages
cat("R using packages package:\n")
```

```
## R using packages package:
```

```
library(survey)
```

```
## Carregando pacotes exigidos: grid
```

```
## Carregando pacotes exigidos: Matrix
```

```
## Carregando pacotes exigidos: survival
```

```
##
```

```
## Anexando pacote: 'survey'
```

```
## O seguinte objeto é mascarado por 'package:graphics':
```

```
##
```

```
## dotchart
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats   1.0.0      v stringr   1.5.1
```

```
## v ggplot2    3.5.1      v tibble    3.2.1
```

```
## v lubridate  1.9.4      v tidyr     1.3.1
```

```
## v purrr      1.0.4
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x tidyr::expand() masks Matrix::expand()
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()    masks stats::lag()
```

```
## x tidyr::pack()   masks Matrix::pack()
```

```
## x tidyr::unpack() masks Matrix::unpack()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(skimr)
```

```
library(nortest)
```

```
library(rio)
```

```
library(here)
```

```
## here() starts at C:/Users/Usuário/Desktop/rscrip/SB_MASLD
```

```
# Display Version Information
```

```
cat("R package versions:\n")
```

```
## R package versions:
```

```
for (p in c("survey", "tidyverse", "skimr", "nortest", "rio", "here")) {  
  cat(p, ": ", as.character(packageVersion(p)), "\n")  
}
```

```
## survey : 4.4.2
## tidyverse : 2.0.0
## skimr : 2.1.5
## nordest : 1.0.4
## rio : 1.2.3
## here : 1.0.1
```

2. Data Selection and Acquisition

Function to download the required survey cycles for a component file

```
downloadNHANES <- function(file, yrs) {
  print(file)
  outdf <- data.frame(NULL)

  for (j in seq_along(file)) {
    urlstring <- paste("wwwn.cdc.gov/Nchs/Data/Nhanes/Public/", yrs[j], "/DataFiles/", file[j], ".XPT",
      download.file(urlstring, tf <- tempfile(), mode = "wb")
    tmpframe <- foreign::read.xport(tf)
    outdf <- bind_rows(outdf, tmpframe)
  }
  return(outdf)
}
```

Specify the survey cycles required, with corresponding file suffixes

Demographic data

```
table_demo <- c("P_DEMO")
```

Examination data

```
table_bpxo <- c("P_BPXO")
table_bmx <- c("P_BMX")
table_lux <- c("P_LUX")
table_fastqx <- c("P_FASTQX")
```

Laboratory Data

```
table_alb_cr <- c("P_ALB_CR")
table_ucpreg <- c("P_UCPREG")
table_hepa <- c("P_HEPA")
table_hepb_s <- c("P_HEPB_S")
table_hepbd <- c("P_HEPBD")
table_hepc <- c("P_HEPC")
table_tchol <- c("P_TCHOL")
table_hdl <- c("P_HDL")
table_glu <- c("P_GLU")
table_ins <- c("P_INS")
table_ghb <- c("P_GHB")
table_trigly <- c("P_TRIGLY")
table_biopro <- c("P_BIOPRO")
```

Questionnaire Data

```

table_bpq <- c("P_BPQ")
table_heq <- c("P_HEQ")
table_paq <- c("P_PAQ")
table_diq <- c("P_DIQ")
table_smq <- c("P_SMQ")
table_dbq <- c("P_DBQ")
table_alq <- c("P_ALQ")
table_mcq <- c("P_MCQ")

```

Dietary data

```

table_dr1tot <- c("P_DR1TOT")
table_dr2tot <- c("P_DR2TOT")

```

Medications

```

table_rxq_rx <- c("P_RXQ_RX")
table_rxq_drug <- c("RXQ_DRUG")

```

Downloading NHANES tables as a temporary files

```

yrs <- c("2017")

```

```

DEMO <- downloadNHANES(table_demo, yrs) # Demographic

```

```

## [1] "P_DEMO"

```

```

BPX <- downloadNHANES(table_bpxo, yrs) # Blood pressure

```

```

## [1] "P_BPX0"

```

```

BMX <- downloadNHANES(table_bmx, yrs) # Body Measures

```

```

## [1] "P_BMX"

```

```

LUX <- downloadNHANES(table_lux, yrs) # Liver Ultrasound Transient Elastography

```

```

## [1] "P_LUX"

```

```

ALB_CR <- downloadNHANES(table_alb_cr, yrs) # Albumin

```

```

## [1] "P_ALB_CR"

```

```

HEPBD <- downloadNHANES(table_hepbd, yrs) # Hepatitis B: Core, Surface antigen, and Hepatitis D

```

```

## [1] "P_HEPBD"

```

```

HEPC <- downloadNHANES(table_hepc, yrs) # Hepatitis C

```

```

## [1] "P_HEPC"

```

```
TCHOL <- downloadNHANES(table_tchol, yrs) # Total cholesterol
```

```
## [1] "P_TCHOL"
```

```
HDL <- downloadNHANES(table_hdl, yrs) # HDL
```

```
## [1] "P_HDL"
```

```
GLU <- downloadNHANES(table_glu, yrs) # Fasting glucose
```

```
## [1] "P_GLU"
```

```
INS <- downloadNHANES(table_ins, yrs) # Insulin
```

```
## [1] "P_INS"
```

```
GHB <- downloadNHANES(table_ghb, yrs) # Glycohemoglobin
```

```
## [1] "P_GHB"
```

```
TRIGLY <- downloadNHANES(table_trigly, yrs) # TG
```

```
## [1] "P_TRIGLY"
```

```
BPQ <- downloadNHANES(table_bpq, yrs) # Hypertension
```

```
## [1] "P_BPQ"
```

```
HEQ <- downloadNHANES(table_heq, yrs) # Hepatitis
```

```
## [1] "P_HEQ"
```

```
PAQ <- downloadNHANES(table_paq, yrs) # Physical activity
```

```
## [1] "P_PAQ"
```

```
DIQ <- downloadNHANES(table_diq, yrs) # Diabetes
```

```
## [1] "P_DIQ"
```

```
SMQ <- downloadNHANES(table_smq, yrs) # Smoke
```

```
## [1] "P_SMQ"
```

```
BIOPRO <- downloadNHANES(table_biopro, yrs) # Standard Biochemistry
```

```
## [1] "P_BIOPRO"
```

```
RXQ_RX <- downloadNHANES(table_rxq_rx, yrs) # Drug use
```

```
## [1] "P_RXQ_RX"
```

```
DBQ <- downloadNHANES(table_dbq, yrs) # Diet
```

```
## [1] "P_DBQ"
```

```
ALQ <- downloadNHANES(table_alq, yrs) # Alcohol
```

```
## [1] "P_ALQ"
```

```
MCQ <- downloadNHANES(table_mcq, yrs) # Medical Conditions - Heart diseases
```

```
## [1] "P_MCQ"
```

```
DR1TOT <- downloadNHANES(table_dr1tot, yrs) # Diet
```

```
## [1] "P_DR1TOT"
```

```
DR2TOT <- downloadNHANES(table_dr2tot, yrs) # Diet_2
```

```
## [1] "P_DR2TOT"
```

```
yrs <- c("1988")
```

```
RXQ_DRUG <- downloadNHANES(table_rxq_drug, yrs) # Drug information
```

```
## [1] "RXQ_DRUG"
```

```
# Merging Tables and Selecting Relevant Variables
```

```
df_temp <- select(DEMO, "SEQN", "SDDSRVYR", "SDMVPSU", "SDMVSTRA", "RIAGENDR", "RIDAGEYR", "RIDRETH1",  
  left_join(., select(BPX, "SEQN", "BPXODI1", "BPXODI2", "BPXODI3", "BPXOSY1", "BPXOSY2", "BPXOSY3"), by = "SEQN"),  
  left_join(., select(BMX, "SEQN", "BMXBMI", "BMXWAIST"), by = "SEQN") %>%  
  left_join(., select(LUX, "SEQN", "LUAXSTAT", "LUAPNME", "LUXSMED", "LUXSIQR", "LUXSIQRM", "LUXCAPM",  
  left_join(., select(ALB_CR, "SEQN", "URDACT"), by = "SEQN") %>%  
  left_join(., select(HEPBD, "SEQN", "LBDHGB"), by = "SEQN") %>%  
  left_join(., select(HEPC, "SEQN", "LBXHCR"), by = "SEQN") %>%  
  left_join(., select(TCHOL, "SEQN", "LBXTC"), by = "SEQN") %>%  
  left_join(., select(HDL, "SEQN", "LBDHDD"), by = "SEQN") %>%  
  left_join(., select(TRIGLY, "SEQN", "WTSAPFRP", "LBDLDL", "LBXTR"), by = "SEQN") %>%  
  left_join(., select(GLU, "SEQN", "LBXGLU"), by = "SEQN") %>%
```

```

left_join(., select(INS, "SEQN", "LBXIN"), by = "SEQN") %>%
left_join(., select(GHB, "SEQN", "LBXGH"), by = "SEQN") %>%
left_join(., select(BPQ, "SEQN", "BPQ020", "BPQ080"), by = "SEQN") %>%
left_join(., select(SMQ, "SEQN", "SMQ020", "SMQ040", "SMQ050Q", "SMQ050U"), by = "SEQN") %>%
left_join(., select(BIOPRO, "SEQN", "LBXSATSI", "LBXSASSI", "LBXSGTSI"), by = "SEQN") %>%
left_join(., select(PAQ, "SEQN", "PAQ605", "PAQ610", "PAD615", "PAQ620", "PAQ625", "PAD630", "PAQ635")
left_join(., select(DIQ, "SEQN", "DIQ010"), by = "SEQN") %>%
left_join(., select(ALQ, "SEQN", "ALQ111", "ALQ121", "ALQ130"), by = "SEQN") %>%
left_join(., select(DR1TOT, "SEQN", "DR1DRSTZ", "DRDINT", "DRQSDIET", "DR1TKCAL", "DR1TPROT", "DR1TCAL")
left_join(., select(DR2TOT, "SEQN", "DR2DRSTZ", "DR2TKCAL", "DR2TPROT", "DR2TCARB", "DR2TSUGR", "DR2T")

# observando o número de linhas e colunas
dim(df_temp)

```

```
## [1] 15560      86
```

```
# glimpse(df_temp)
```

3. Data Cleaning and Preprocessing

```
# Specifying variable types
```

```

df_temp <- df_temp %>%
  mutate(across(c(SEQN, SDDSRVYR, RIDAGEYR, BPXODI1, BPXODI2, BPXODI3, BPXOSY1, BPXOSY2, BPXOSY3, LBXTOT,
  mutate(across(c(RIAGENDR, RIDRETH1, DMDEDUC2, RIDEXPRG, LUAXSTAT, LUAPNME, LBDHBG, LBXHCR, BPQ020, BPQ080,

```

```
# Recode Values
```

```

df_temp$RIDRETH1 <- recode_factor(
  as.factor(df_temp$RIDRETH1),
  "1" = 1,
  "2" = 1,
  "3" = 3,
  "4" = 4,
  "5" = 5
)

```

```

df_temp$RIDRETH3 <- recode_factor(
  as.factor(df_temp$RIDRETH1),
  "1" = 1,
  "2" = 1,
  "3" = 3,
  "4" = 4,
  "6" = 6,
  "7" = 7
)

```

```

df_temp$DMDEDUC2 <- recode_factor(df_temp$DMDEDUC2,
  "1" = "1",
  "2" = "1",

```

```

"3" = "2",
"4" = "3",
"5" = "4",
"7" = NA_character_,
"9" = NA_character_
)

```

Notes: INDFMPIR - The value was not computed if the respondent only reported income as < \$20,000 or \$20,000. There is 2201 missing value na tabela geral. RIDEXPRG: Because of possible disclosure risks, pregnancy status is only released for women aged 20-44 years.

RIDEXPRG: Pregnancy status at the time of the health examination was ascertained for females 8–59 years of age. Due to disclosure risks pregnancy status is only released for women 20-44 years of age. The information used to code RIDEXPRG values included self-reported pregnancy status and urine pregnancy test results. Persons who reported they were pregnant at the time of exam were assumed to be pregnant (RIDEXPRG=1). Those who reported they were not pregnant or did not know their pregnancy status were further classified based on the results of the urine pregnancy test. If the respondent reported “no” or “don’t know” and the urine test result was positive, the respondent was coded as pregnant (RIDEXPRG=1). If the respondent reported “no” and the urine test was negative, the respondent was coded not pregnant (RIDEXPRG=2). If the respondent reported did not know her pregnancy status and the urine test was negative, the respondent was coded “could not be determined” (RIDEXPRG=3). Persons who were interviewed, but not examined also have an RIDEXPRG value = 3 (could not be determined). The percentage of pregnant women/girls aged 8-19 or 45-59 years is less than 1% in the 2017-March 2020 pre-pandemic sample.

Notes: - BPX - Systolic BP cannot be greater than 300 mmHg; Systolic blood pressure must be greater than diastolic BP; If there is no systolic BP, there can be no diastolic BP. However, there can be a systolic measurement without a diastolic measurement.

Existem valores que poderiam ser considerados irrealis, e não existe nada na documentação que informe que esses valores foram revisados. Com exceção das regras acima. Por isso é Necessário estabelecer limites razoáveis para PAD (Por exemplo, 40 - 120 mmHg). Verificar quais valores são utilizados na literatura

VERIFICAR - We excluded all of these subjects, and indeed any subject who had at least one measurement recorded outside the ranges (40,140) for diastolic and (60,250) for systolic BP, #as recommended by the CDC (Littman et al. 2012) : NHANES Blood Pressure-Based Mortality Risk - Appendix

NOTES All participants 12 years and older in the NHANES 2017-March 2020 pre-pandemic sample were eligible. Participants were excluded if they (1) were unable to lie down on the exam table, (2) were pregnant (or unsure if pregnant) at the time of their exam, or a urine could not be obtained to test for pregnancy, (3) had an implanted electronic medical device, or (4) were wearing a bandage or had lesions on the right side of their abdomen by the ribs (where measurements would be taken).

The elastography exam was performed by NHANES health technicians (HTs), who were trained and certified by NHANES staff and the equipment manufacturer (Echosens™ North America). The exams were performed according to the manufacturer guidelines. To help maintain a standardized data quality, the machine conducts and displays several quality control (QC) measures during the test: 1) the median of all valid measurements performed during the examination; 2) interquartile range (IQR) which represents the interval around the median within which 50% of all valid measurements will fall; and 3) IQR/M: the ratio of the IQR to the median stiffness. All these QC indexes were recalculated after each new measurement. HTs were trained to take 10 valid measurements with an IQR/M ratio less than 30%.

If the first 10 measurements taken had an IQR/M 30%, the HT may choose to capture additional measurements until the IQR/M index was lower than 30%. The HT also has the option to delete outlier measurements to lower the IQR/M from the list of valid measurements. It should be noted that HTs were only allowed to delete measurements from the beginning of the measurements. Once a measurement was chosen to be deleted, any measurements taken before the chosen one will be eliminated from the exam as well. This is to reduce bias, so HTs were unable to hand select which measurement(s) to delete.

The QC criteria of IQR/M <30% is to reduce variability and improve validity by taking measurements that result in few outliers. Multiple factors can affect the measurement and result in outliers, for example, a participant moves during the exam, or technical issues, such as the placement of the probe is not centered over the liver or perpendicular, or if structures such as lung or ribs appear while the measurement is captured. Participants with a lot of adipose tissue sometimes may make it difficult for HTs to capture 10 valid measurements. Because of the twelve-minute time limit and the challenge of locating the proper site to administer the exam on the body, HTs could end up retaining fewer than 10 valid measurements.

The liver elastography exam status code (LUAXSTAT) was created and indicates the following for each participant: 1 = Complete exam (i.e., fasting time of at least 3 hours, 10 or more complete stiffness (E) measures, and a liver stiffness interquartile (IQR_E) range / median E < 30%). 2 = Partial exam (i.e., either a fasting time < 3 hours, < 10 complete stiffness (E) measures, or a liver stiffness interquartile (IQR_E) range / median E 30% or higher). 3 = Ineligible participant (see eligibility criteria above). 4 = Not done (i.e., refusal, limited time during exam visit, other).

```
df_temp <- df_temp %>%

# Exclusion of laboratory data for individuals without adequate fasting or without a blood sample
# WTSAFPRP == 0 indicates inadequate fasting or absence of blood sample.
# In these situations, the biomarkers LBXTR, LBDLDL, LBXGLU, and LBXIN are treated as NA.
mutate(
  LBXTR = ifelse(WTSAFPRP == 0, NA, LBXTR),
  LBDLDL = ifelse(WTSAFPRP == 0, NA, LBDLDL),
  LBXGLU = ifelse(WTSAFPRP == 0, NA, LBXGLU),
  LBXIN = ifelse(WTSAFPRP == 0, NA, LBXIN)
) %>%

# Exclusão de respostas inválidas nas variáveis de atividade física (recusa ou desconhecimento)
mutate(
  PAQ610 = ifelse(PAQ610 %in% c(77, 99), NA, PAQ610),
  PAD615 = ifelse(PAD615 %in% c(7777, 9999), NA, PAD615),
  PAQ620 = ifelse(PAQ620 %in% c(7, 9), NA, PAQ620),
  PAQ625 = ifelse(PAQ625 %in% c(77, 99), NA, PAQ625),
  PAQ640 = ifelse(PAQ640 %in% c(77, 99), NA, PAQ640),
  PAD630 = ifelse(PAD630 %in% c(7777, 9999), NA, PAD630),
  PAD645 = ifelse(PAD645 %in% c(7777, 9999), NA, PAD645),
  PAQ655 = ifelse(PAQ655 %in% c(77, 99), NA, PAQ655),
  PAD660 = ifelse(PAD660 %in% c(7777, 9999), NA, PAD660),
  PAQ670 = ifelse(PAQ670 %in% c(77, 99), NA, PAQ670),
  PAD675 = ifelse(PAD675 %in% c(7777, 9999), NA, PAD675)
) %>%

# Exclusão de respostas inválidas para tempo sentado (recusa ou desconhecimento)
mutate(
  PAD680 = ifelse(PAD680 %in% c(7777, 9999), NA, PAD680)
) %>%

# Exclusão de respostas inválidas para consumo de álcool (recusou ou não soube responder)
mutate(
  ALQ130 = ifelse(ALQ130 %in% c(777, 999), NA, ALQ130)
) %>%

# Exclusão de respostas inválidas para fumo (recusou ou não soube responder)
mutate(
```

```

    SMQ020 = ifelse(SMQ020 %in% c(7, 9), NA, SMQ020)
  ) %>%

  # Exclusão de dados dietéticos não confiáveis ou com ingestão energética fora do intervalo plausível
  mutate(across(
    c("DR1TPROT", "DR1TCARB", "DR1TSUGR", "DR1TTFAT", "DR1TSFAT", "DR1TMFAT", "DR1TPFAT",
      "DR2TPROT", "DR2TCARB", "DR2TSUGR", "DR2TTFAT", "DR2TSFAT", "DR2TMFAT", "DR2TPFAT"),
    ~ ifelse(DR1DRSTZ != 1 | DR2DRSTZ != 1 | DR1TKCAL < 500 | DR1TKCAL > 3500 | DR2TKCAL < 500 | DR2TKCAL > 3500, NA, .x)
  )) %>%

  # Exclusão de respostas inválidas para diagnóstico de diabetes autoreportado (Borderline/Prédiabetes)
  mutate(
    DIQ010 = ifelse(DIQ010 %in% c(7, 9), NA, DIQ010),
    DIQ010 = ifelse(DIQ010 == 3, 2, DIQ010)
  ) %>%

  # Exclusão de respostas inválidas para diagnóstico de HIPERTENSÃO autoreportado
  mutate(
    BPQ020 = ifelse(BPQ020 %in% c(7, 9), NA, BPQ020),
  )

```

P_BMX - The P_BMX data were reviewed for unusual and erroneous values. Review criteria were based on the NHANES 1999–2016 body measurement data. During the data review, values that were above the 99th percentile or below the 1st percentile, for a specific age or age-gender group, were flagged for review. When records were flagged, the entire body measures record was reviewed for reasonableness. Subject characteristics, such as height, weight, age, and gender were taken into consideration. **Values that were determined to be unrealistic were deleted from the file.** None of the original body measures data were changed and there are no imputed values in this file.

ALB_CR, P_HEPBD, P_HEPC, TCHOL, P_HDL, P_TRIGLY, P_GLU, P_INS, P_GLU, P_BIOPRO - The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

P_TRIGLY, P_GLU, P_INS - Measured only in a fasting subsample of participants 12 years and older. Special sample weights are required to analyze these data properly. Participants included in the fasting subsample but did not provide a blood specimen (n=346) have an assigned sample weight value of “0” in their records. In addition, participants whom provided a blood specimen but did not meet the 8 to less than 24 hours fasting criteria (n=268) have the sample weight value assigned as “0” (WTSAFPRP =0) as well. Six of these 268 participants did not provide enough volume to have their triglycerides tested. Triglyceride measurements for the other 262 participants are included in the dataset. However, to include these data in the analysis, a reweighting is required. There are another 88 participants in the fasting subsample that did not provide enough blood; therefore, they did not have triglycerides measured. However, these participants have data in at least one other fasting subsample tests (e.g., plasma fasting glucose). Therefore, they have a sample weight larger than “0,” regardless of missing their triglyceride test results. *Missing data for TRIGLY is very high since it was only measured in a fasting subsample.*

P_BPQ, P_DIQ - After collection, interview data were reviewed by the NHANES field office staff for accuracy and completeness of selected items. The interviewers were required to audio-record interviews and the recorded interviews were reviewed by NCHS staff and interviewer supervisors. Frequency counts were checked, “skip” patterns were verified, and the reasonableness of question responses was reviewed. Edits were made to some variables to ensure the completeness, consistency, and analytic usefulness of the data. Edits were also made, when necessary, to address data disclosure concerns. *Although the BPQ data were collected as part of the household questionnaire, if they are merged with the MEC exam data, exam sample weights should be used for the analyses. Refused and don't know foram mantidos e de acordo com a equação de classificação foram classificados como não (se exames normais)*

P_BIOPRO - NHANES não recomenda a utilização dos dados de glicose, colesterol total e triglicerídeos que constam nessa tabela. Em vez disso, deve-se utilizar os dados das tabelas P_GLU, TCHOL e P_TRIGLY.

P_HEPBC - *Anti-HBc positivo (LBXHBC) = exposição passada ou atual ao HBV. Devemos utilizar?*

HEPB_S - The hepatitis B surface antibody test is used in NHANES to assess immunity due to vaccination. Thus, we do not use this variable.

P_HEPC - Samples found reactive to the HCV antibody screening test are tested for HCV RNA. *Anti-HCV positivo (LBDHCI) = exposição ao vírus (presente ou passada), Devemos utilizar*

P_HEQ - Questionário sobre infecção passada por hepatite B e C. ~ 100 responderam sim para cada uma dessas doenças. Não irei utilizar pelo risco de viés que o questionário introduz. Além disso, temos os dados laboratoriais.

P_SMQ - Edits were made to ensure completeness, consistency, and analytic usefulness of the data. A variable, SMAQUX, was created as a flag to designate each interview mode and age group.

P_PAQ - After collection, interview data were reviewed by the NHANES field office staff for accuracy and completeness of selected items. The interviewers were required to audio-record interviews and the recorded interviews were reviewed by NCHS staff and interviewer supervisors. PAQ foi substituído por valores que correspondiam a “refused” ou “don’t know” por NA.

P_ALQ - The data were edited for completeness, consistency, and illogical values. Because the data are based on self-reports, not all inconsistencies could be resolved. It is recommended that analysts review the data carefully prior to analysis.

Participants were considered eligible for inclusion in this study if the following criteria were met: 1) adult patients (age ≥ 18 years old); 2) having a complete transient elastography (TE) exam, i.e., fasting time of at least 3 h, 10 or more complete stiffness (E) measures, and a liver stiffness interquartile range (IQR) to median E ratio <30%; and 3) having body mass index (BMI) data. We excluded patients who met any of the following conditions: 1) BMI <18.5 kg/m² or >25 kg/m² (>23 kg/m² for patients of Asian descent); 2) heavy alcohol use, defined as >2 drinks per day for men and >1 per day for women; 3) positive hepatitis C (HCV) RNA; 4) positive hepatitis B surface antigen (HBsAg); or 5) exposure to steatogenic drugs, such as tamoxifen, valproic acid, amiodarone, methotrexate, and glucocorticoids.

<https://pmc.ncbi.nlm.nih.gov/articles/PMC10662062/>

3.1 Construction of Derived Variables

```
### MEDICATION USE

# Defining medications of interest

antihyperlipidemic_agents <- RXQ_DRUG %>%
  filter(RXDDCN1B == "ANTHYPERLIPIDEMIC AGENTS") %>%
  distinct(RXDDRUG) %>%
  arrange(RXDDRUG)

# Filtrar medicamentos da classe "ADRENAL CORTICAL STEROIDS" + fármacos específicos
nafld_extra_pattern <- "(amiodarone|methotrexate|tamoxifen|valproic acid)"

nafld_induced_agents <- RXQ_DRUG %>%
  filter(
    RXDDCN1B == "ADRENAL CORTICAL STEROIDS" |
    grepl(nafld_extra_pattern, RXDDRUG, ignore.case = TRUE)
  ) %>%
```

```

distinct(RXDDRUG) %>%
arrange(RXDDRUG)

# Apply patterns and summarize by participant

medication_flags <- RXQ_RX %>%
  mutate(
    LIPID_LOWERING_AGENTS_FLAG = grepl(paste(antihyperlipidemic_agents$RXDDRUG, collapse = "|"), RXQ_RX)
    NAFLD_INDUCED_AGENTS_FLAG = grepl(paste(naflld_induced_agents$RXDDRUG, collapse = "|"), RXQ_RX)
  ) %>%
  group_by(SEQN) %>%
  summarise(
    LIPID_LOWERING_AGENTS = any(LIPID_LOWERING_AGENTS_FLAG),
    NAFLD_INDUCED_AGENTS = any(NAFLD_INDUCED_AGENTS_FLAG),
    .groups = "drop"
  )

df_temp <- df_temp %>%
  left_join(medication_flags, by = "SEQN") %>%

  mutate(

    ### PRESSÃO ARTERIAL -----
    SBP = rowMeans(select(., BPXOSY1, BPXOSY2, BPXOSY3), na.rm = TRUE),
    DBP = rowMeans(select(., BPXODI1, BPXODI2, BPXODI3), na.rm = TRUE)) %>%

  mutate(
    ### CLASSIFICAÇÃO SOBREPESO OU OBESIDADE / HIGH WAIST CIRCUMFERENCE -----
    OVERWEIGHT_OBESE = as.factor(case_when(
      is.na(BMXBMI) & is.na(BMXWAIST) ~ NA,

      # Asiáticos
      (RIDRETH3 == 6 & BMXBMI >= 23) |
      (RIDRETH3 == 6 & RIAGENDR == 1 & BMXWAIST >= 90) |
      (RIDRETH3 == 6 & RIAGENDR == 2 & BMXWAIST >= 80) ~ 1,

      # Não asiáticos
      (RIDRETH3 != 6 & BMXBMI >= 25) |
      (RIDRETH3 != 6 & RIAGENDR == 1 & BMXWAIST >= 94) |
      (RIDRETH3 != 6 & RIAGENDR == 2 & BMXWAIST >= 80) ~ 1,

      TRUE ~ 0
    )),

    ### HYPERTRIGLICERIDEMIA CLASSIFICATION -----

    HYPERTRIGLY = as.factor(case_when(
      is.na(LBXTR) & (LIPID_LOWERING_AGENTS == FALSE) ~ NA,
      LBXTR >= 150 ~ 1,
      LIPID_LOWERING_AGENTS == TRUE ~ 1,
      TRUE ~ 0)),

    ### LOW-HDL CLALSSIFICATION -----

```

```

LOW_HDL = as.factor(case_when(
  RIAGENDR == 1 & LBDHDD < 40 ~ 1,
  RIAGENDR == 2 & LBDHDD < 50 ~ 1,
  LBDHDD >= 40 & RIAGENDR == 1 ~ 0, # Para homens com HDL normal (>= 40)
  LBDHDD >= 50 & RIAGENDR == 2 ~ 0,
  TRUE ~ NA
)),

### DIABETES CLASSIFICATIONS -----
DIABETES = as.factor(case_when(
  rowSums(select(., c(DIQ010, LBXGH, LBXGLU)) %>% is.na()) == 3 ~ NA,
  !is.na(DIQ010) & DIQ010 == 1 ~ 1,
  !is.na(LBXGH) & LBXGH >= 6.5 ~ 1,
  !is.na(LBXGLU) & LBXGLU >= 126 ~ 1,
  TRUE ~ 0
)),

### DYSGLYCAEMIA OR TYPE 2 DIABETES CLASSIFICATIONS

DYSGLYCAEMIA_DM2 = as.factor(case_when(
  rowSums(select(., c(DIQ010, LBXGH, LBXGLU)) %>% is.na()) == 3 ~ NA,
  !is.na(DIQ010) & DIQ010 == 1 ~ 1,
  !is.na(LBXGH) & LBXGH >= 5.7 ~ 1,
  !is.na(LBXGLU) & LBXGLU >= 100 ~ 1,
  TRUE ~ 0
)),

### HYPERTENSION CLASSIFICATION -----
HTN = as.factor(case_when(
  rowSums(select(., c(BPQ020, SBP, DBP)) %>% is.na()) == 3 ~ NA,
  !is.na(BPQ020) & BPQ020 == 1 ~ 1,
  !is.na(SBP) & SBP >= 130 ~ 1,
  !is.na(DBP) & DBP >= 85 ~ 1,
  TRUE ~ 0)),

### PHYSICAL ACTIVITY CLASSIFICATION -----
VIGOROUS_PA =
  ifelse(PAQ605 == 1, (PAQ610 * PAD615), 0) +
  ifelse(PAQ650 == 1, (PAQ655 * PAD660), 0),

MODERATE_PA =
  ifelse(PAQ620 == 1, (PAQ625 * PAD630), 0) +
  ifelse(PAQ635 == 1, (PAQ640 * PAD645), 0) +
  ifelse(PAQ665 == 1, (PAQ670 * PAD675), 0),

MVPA = (VIGOROUS_PA * 2) + MODERATE_PA, # Moderate-to-vigorous physical activity
PA_MET = (VIGOROUS_PA * 8) + (MODERATE_PA * 4), # METs

WORK_ACTIVITY =
  ifelse(PAQ605 == 1, (PAQ610 * PAD615), 0) +
  ifelse(PAQ620 == 1, (PAQ625 * PAD630), 0),

TRANSPORTATION_ACTIVITY =
  ifelse(PAQ635 == 1, (PAQ640 * PAD645), 0),

```

```

RECREATIONAL_ACTIVITY =
  ifelse(PAQ650 == 1, (PAQ655 * PAD660), 0) +
  ifelse(PAQ665 == 1, (PAQ670 * PAD675), 0),

### DIET (Mean of 2 days) -----
KCAL_MEAN = rowMeans(select(., DR1TKCAL, DR2TKCAL), na.rm = TRUE),
TPROT_MEAN = rowMeans(select(., DR1TPROT, DR2TPROT), na.rm = TRUE),
TCARB_MEAN = rowMeans(select(., DR1TCARB, DR2TCARB), na.rm = TRUE),
TSUGAR_MEAN = rowMeans(select(., DR1TSUGR, DR2TSUGR), na.rm = TRUE),
TFAT_MEAN = rowMeans(select(., DR1TTFAT, DR2TTFAT), na.rm = TRUE),
TSFAT_MEAN = rowMeans(select(., DR1TSFAT, DR2TSFAT), na.rm = TRUE),
TMFAT_MEAN = rowMeans(select(., DR1TMFAT, DR2TMFAT), na.rm = TRUE),
TPFAT_MEAN = rowMeans(select(., DR1TPFAT, DR2TPFAT), na.rm = TRUE),
TUNSATFAT_MEAN = TMFAT_MEAN + TPFAT_MEAN,

### CLASSIFICAÇÃO DE ÁLCOOL -----

# Heavy drinker > 21g / day in men or > 14g/ day in woman
ALCOHOL = as.factor(case_when(
  ALQ111 == 2 ~ "Non-heavy Drinker",
  !is.na(ALQ130) & ALQ130 > 4 & RIAGENDR == 1 ~ "Heavy Drinker",
  !is.na(ALQ130) & ALQ130 > 3 & RIAGENDR == 2 ~ "Heavy Drinker",
  !is.na(ALQ130) & ALQ130 <= 4 & RIAGENDR == 1 ~ "Non-heavy Drinker",
  !is.na(ALQ130) & ALQ130 <= 3 & RIAGENDR == 2 ~ "Non-heavy Drinker",

  TRUE ~ NA_character_
)),
# Alcohol use
ALCOHOL_USE = as.factor(case_when(
  ALQ111 == 2 ~ 0,
  !is.na(ALQ130) & ALQ130 < 2 & RIAGENDR == 1 ~ 1,
  !is.na(ALQ130) & ALQ130 < 1 & RIAGENDR == 2 ~ 1,
  !is.na(ALQ130) & ALQ130 >= 2 & RIAGENDR == 1 ~ 2,
  !is.na(ALQ130) & ALQ130 >= 1 & RIAGENDR == 2 ~ 2,

  TRUE ~ NA
)),

### SMOKING -----
SMOKING = as.factor(case_when(
  SMQ020 == 2 ~ 0,
  SMQ040 %in% c(1,2) ~ 2,
  SMQ040 == 3 ~ 1,
  TRUE ~ NA
)),

### ÍNDICES CLÍNICOS -----
AST_ALT = LBXSATSI / LBXSASSI,
HOMAIR = (LBXIN * LBXGLU) / 405, # Matthews et al., Diabetologia 1985

HSI = (8 * LBXSATSI / LBXSASSI) + BMXBMI +
  ifelse(RIAGENDR == 2, 2, 0) + # +2 se mulher
  ifelse(DIABETES == "Yes", 2, 0), # HSI: Lee JH et al., Hepatology 2010

```

```

FLI_LOGIT = 0.953 * log(LBXTR) +
            0.139 * BMXBMI +
            0.718 * log(LBXSGTSSI) +
            0.053 * BMXWAIST - 15.745,

FLI = exp(FLI_LOGIT) / (1 + exp(FLI_LOGIT)) # FLI: Bedogni et al., BMC Gastroenterol 2006
)

```

#NOTA: HDL VALOR ABAIXO DO PONTO DE CORTE OU USO DE HIPOLIPEMIANTE?

Declarando valores

RIAGENDER = 1 = male RIAGENDER = 2 = female

DIQ010 = 1 = YES for Doctor told you have diabetes; DIQ050 = 1 = YES for Taking insulin now; DIQ070 = 1 = YES for Take diabetic pills to lower blood sugar; *Diabetes was defined as glycated hemoglobin (HbA1c) 6.5%, fasting plasma glucose 7.0 mmol/L (126 mg/dL), or previous diabetes diagnosis following current American Diabetes Association guidelines. (<https://doi.org/10.1210/clinem/dgab762>)*

No R, a função log() sem especificar a base calcula o logaritmo natural (ln), ou seja, log na base e. Da mesma forma, exp() representa a exponenciação pela constante de Euler (e 2.7182818)."

Adjustment of nutrient intake by energy intake (Willet 1998)
Willet_adjustment - the function takes the values of nutrient intake and energy intake, and returns t

```

willet_adjustment <- function(a, b) {
  df <- data.frame(a, b)
  lm <- lm(a ~ b)
  df$nutrient_predicted <- lm$coefficients[[1]] + lm$coefficients[[2]] * df[, 2]
  df$nutrient_residual <- df[, 1] - df$nutrient_predicted
  df$nutrient_constant <- lm$coefficients[[1]] + lm$coefficients[[2]] * mean(df[, 2], na.rm = TRUE)
  df$nutrient_adj <- df$nutrient_constant + df$nutrient_residual
  residuals_normality <- ad.test(lm$residuals)
  +if (residuals_normality$p.value < 0.05) {
    return(df$nutrient_adj)
  } else {
    return("normality of residuals not achieved")
  }
}

```

Função para plotar e adicionar o valor de r

```

plot_with_correlation <- function(x, y, x_name, y_name) {
  plot(x, y, col = "lightblue", xlab = x_name, ylab = y_name)
  abline(lm(y ~ x), col = "red", lwd = 3)
  correlation <- cor(x, y, use = "pairwise.complete.obs") # Calcula a correlação
  text(min(x, na.rm = TRUE), max(y, na.rm = TRUE), # Posiciona o texto no canto superior esquerdo
       labels = paste("r =", round(correlation, 2)),
       adj = c(0, 1), # Ajusta a posição do texto
       cex = 0.8,     # Tamanho do texto
       col = "blue")
}

```

Adjustment for Energy Intake

```

df_temp$TPROT_ADJ <- willet_adjustment(df_temp$TPROT_MEAN, df_temp$KCAL_MEAN)
df_temp$TCARB_ADJ <- willet_adjustment(df_temp$TCARB_MEAN, df_temp$KCAL_MEAN)
df_temp$TSUGAR_ADJ <- willet_adjustment(df_temp$TSUGAR_MEAN, df_temp$KCAL_MEAN)
df_temp$TFAT_ADJ <- willet_adjustment(df_temp$TFAT_MEAN, df_temp$KCAL_MEAN)
df_temp$TSFAT_ADJ <- willet_adjustment(df_temp$TSFAT_MEAN, df_temp$KCAL_MEAN)
df_temp$TMFAT_ADJ <- willet_adjustment(df_temp$TMFAT_MEAN, df_temp$KCAL_MEAN)
df_temp$TPFAT_ADJ <- willet_adjustment(df_temp$TPFAT_MEAN, df_temp$KCAL_MEAN)
df_temp$TUNSATFAT_ADJ <- willet_adjustment(df_temp$TUNSATFAT_MEAN, df_temp$KCAL_MEAN)

```

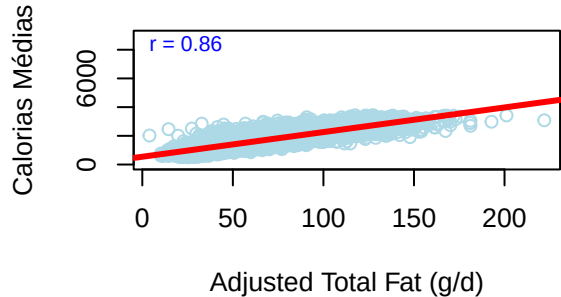
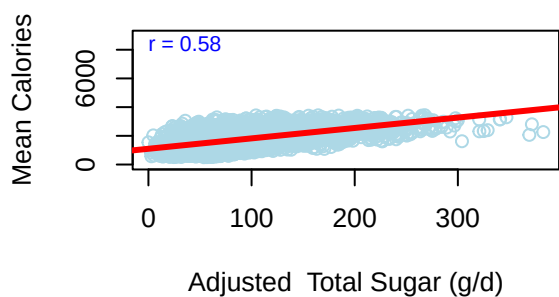
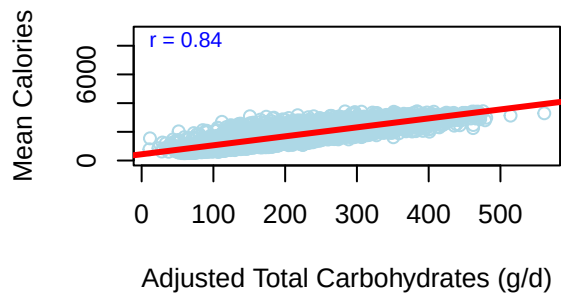
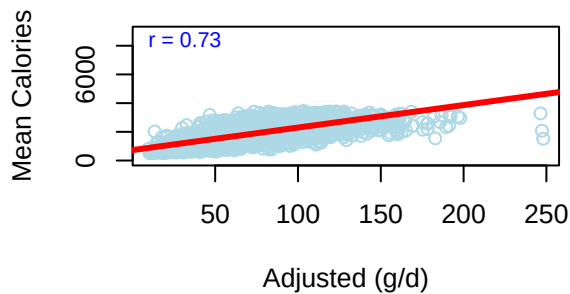
Checking Correlation Before Adjustment for Energy Intake

```
par(mfrow = c(2, 2))
```

```

plot_with_correlation(df_temp$TPROT_MEAN, df_temp$KCAL_MEAN, "Adjusted (g/d)", "Mean Calories")
plot_with_correlation(df_temp$TCARB_MEAN, df_temp$KCAL_MEAN, "Adjusted Total Carbohydrates (g/d)", "Mean Calories")
plot_with_correlation(df_temp$TSUGAR_MEAN, df_temp$KCAL_MEAN, "Adjusted Total Sugar (g/d)", "Mean Calories")
plot_with_correlation(df_temp$TFAT_MEAN, df_temp$KCAL_MEAN, "Adjusted Total Fat (g/d)", "Calorias Médias")

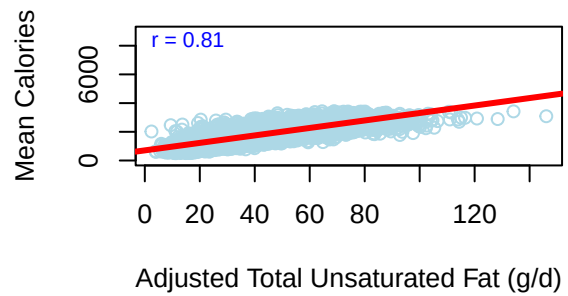
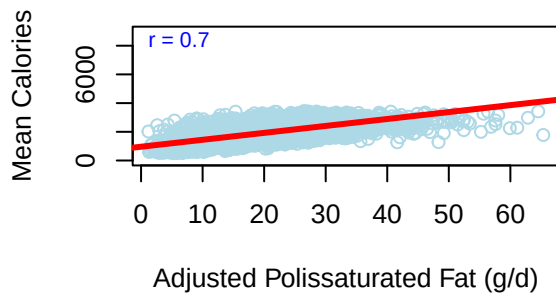
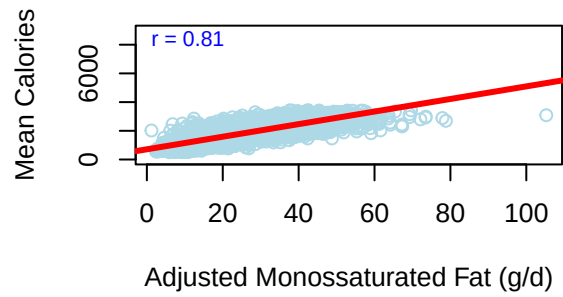
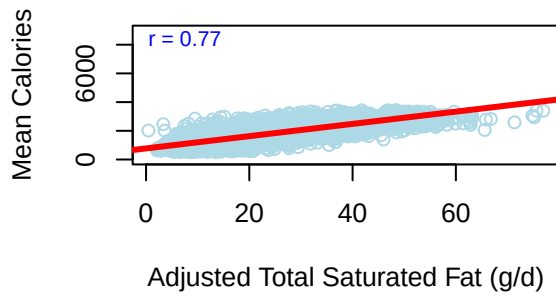
```



```

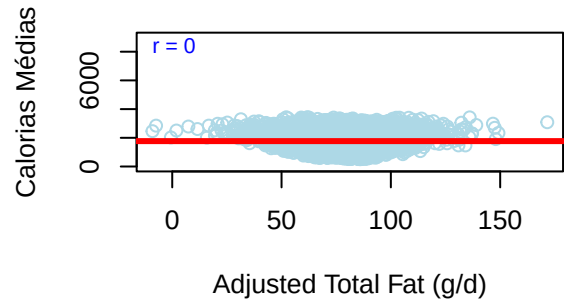
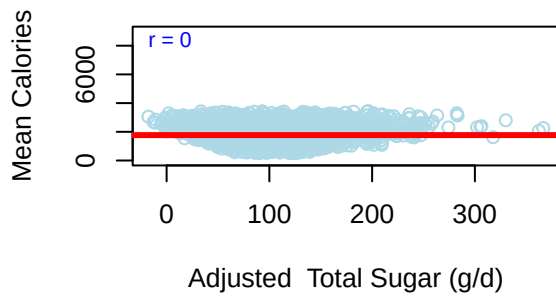
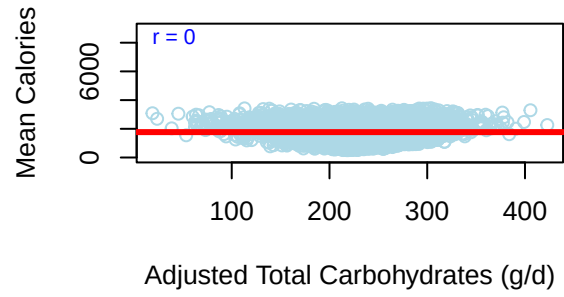
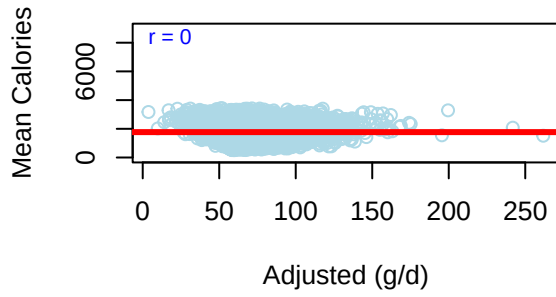
plot_with_correlation(df_temp$TSFAT_MEAN, df_temp$KCAL_MEAN, "Adjusted Total Saturated Fat (g/d)", "Mean Calories")
plot_with_correlation(df_temp$TMFAT_MEAN, df_temp$KCAL_MEAN, "Adjusted Monossaturated Fat (g/d)", "Mean Calories")
plot_with_correlation(df_temp$TPFAT_MEAN, df_temp$KCAL_MEAN, "Adjusted Polissaturated Fat (g/d)", "Mean Calories")
plot_with_correlation(df_temp$TUNSATFAT_MEAN, df_temp$KCAL_MEAN, "Adjusted Total Unsaturated Fat (g/d)", "Mean Calories")

```

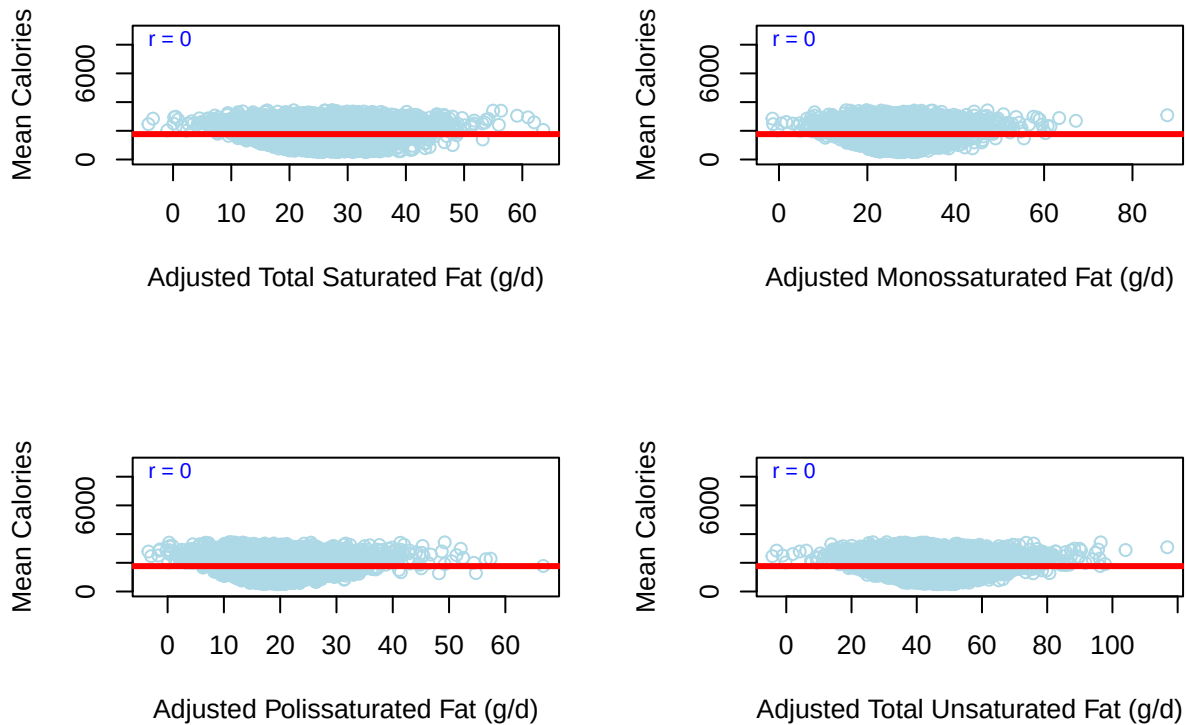


Checking Correlation After Adjustment for Energy Intake. This correlation should be close to 0.

```
plot_with_correlation(df_temp$TPROT_ADJ, df_temp$KCAL_MEAN, "Adjusted (g/d)", "Mean Calories")
plot_with_correlation(df_temp$TCARB_ADJ, df_temp$KCAL_MEAN, "Adjusted Total Carbohydrates (g/d)", "Mean Calories")
plot_with_correlation(df_temp$TSUGAR_ADJ, df_temp$KCAL_MEAN, "Adjusted Total Sugar (g/d)", "Mean Calories")
plot_with_correlation(df_temp$TFAT_ADJ, df_temp$KCAL_MEAN, "Adjusted Total Fat (g/d)", "Calorias Médias")
```



```
plot_with_correlation(df_temp$TSFAT_ADJ, df_temp$KCAL_MEAN, "Adjusted Total Saturated Fat (g/d)", "Mean
plot_with_correlation(df_temp$TMFAT_ADJ, df_temp$KCAL_MEAN, "Adjusted Monossaturated Fat (g/d)", "Mean
plot_with_correlation(df_temp$TPFAT_ADJ, df_temp$KCAL_MEAN, "Adjusted Polissaturated Fat (g/d)", "Mean
plot_with_correlation(df_temp$TUNSATFAT_ADJ, df_temp$KCAL_MEAN, "Adjusted Total Unsaturated Fat (g/d)",
```



3.2.1 Criating MASLD and Sedentary Behaviour Groups

```
df_temp <- df_temp %>%
  mutate(NAFLD = as.factor(ifelse(df_temp$LUXCAPM < 248, 0, 1))) %>%
  mutate(MASLD = as.factor(case_when(
    NAFLD == 1 & (
      # Critérios de disfunção cardiometabólica
      OVERWEIGHT_OBESE == 1 |
      HYPERTRIGLY == 1 |
      LOW_HDL == 1 |
      DIABETES == 1 |
      HTN == 1
    ) ~ 1, TRUE ~ 0
  ))) %>%
  mutate(HEPATIC_FIBROUSIS = as.factor(ifelse(df_temp$LUXSMED < 8, 0, 1))) %>%
  mutate(SB_CATEGORIC = as.factor(ntile(df_temp$PAD680, 5)))
```

****_Note_1:** MASLD diagnostic = CAMP ≥ 248 (Reference: EASL–EASD–EASO Clinical Practice Guidelines on the management of metabolic dysfunction-associated steatotic liver disease (MASLD): Executive Summary) **_Note_1**:** Liver stiffness (marker of liver fibrosis) ≥ 8 (Reference: EASL–EASD–EASO Clinical Practice Guidelines on the management of metabolic dysfunction-associated steatotic liver disease (MASLD): Executive Summary)

4. Creating the Working Data Frame

```
# Selecting working variables
```

```
df <- df_temp %>%  
  select(  
    SEQN, WTSAPFRP,SDMVPSU, RIAGENDR, SDMVSTRA, RIDAGEYR, RIDRETH1, RIDRETH3, DMDEDUC2, INDFMPIR, BMXWA
```

```
# Filtrando apenas para garantir pesos válidos (obrigatório antes do svydesign)
```

```
df <- df %>%  
  filter(  
    !is.na(WTSAPFRP), # Peso amostral não pode ser NA  
    WTSAPFRP > 0 # Peso amostral deve ser positivo  
  )
```

```
# Before
```

```
dim(df_temp)
```

```
## [1] 15560 132
```

```
# After
```

```
dim(df)
```

```
## [1] 4476 61
```

```
# Criando o objeto de amostra complexa
```

```
svy_design <- svydesign(  
  ids = ~ SDMVPSU, # Substitua "SDMVPSU" pela sua variável de cluster (se for diferente)  
  strata = ~ SDMVSTRA, # Substitua "SDMVSTR" pela sua variável de estratificação (se for diferente)  
  weights = ~ WTSAPFRP, # Substitua "WTMEC2YR" pela sua variável de peso amostral (se for diferente)  
  data = df, # Seu conjunto de dados  
  nest = TRUE  
)
```

```
svy_design_subset <- subset(svy_design,  
  RIDAGEYR >= 20 &  
  LUAXSTAT == 1 &  
  !is.na(PAD680) &  
  !is.na(LUXCAPM) &  
  (RIDEXPRG != 1 | is.na(RIDEXPRG)) &  
  ALCOHOL != "Heavy drinker" &  
  (LBDHBG != 1 | is.na(LBDHBG)) &  
  (LBXHCR != 1 | is.na(LBXHCR)) &  
  NAFLD_INDUCED_AGENTS == FALSE  
)
```

```
svy_design_subset$variables <- svy_design_subset$variables %>%  
  select(-c(NAFLD_INDUCED_AGENTS, LUAXSTAT, RIDEXPRG, LBDHBG, LBXHCR))
```

```
# Fazer o Fluxograma
```

```
save(svy_design_subset, file = "df.RData")
```